discusses in more detail, this unsustainable reliance on fossil fuels is changing Earth's climate and increasing the amount of waste that humans produce. Ultimately, the combination of resource use per person and population density determines our global ecological footprint.

What factors will eventually limit the growth of the human population? Perhaps food will be the main limiting factor. Malnutrition and famine are common in some regions, but they result mainly from the unequal distribution of food rather than from inadequate production. So far, technological improvements in agriculture have allowed food supplies to keep up with global population growth. In contrast, the demands of many populations have already far exceeded the local and even regional supplies of one renewable resource—fresh water. More than 1 billion people do not have access to sufficient water to meet their basic sanitation needs. The human population may also be limited by the capacity of the environment to absorb its wastes. If so, then Earth's current human occupants could lower the planet's long-term carrying capacity for future generations.

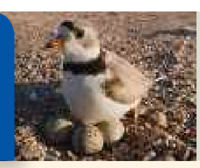
Technology has substantially increased Earth's carrying capacity, but no population can grow indefinitely. After reading this chapter, you should realize that there is no single carrying capacity. How many people our planet can sustain depends on the quality of life each of us has and the distribution of wealth across people and nations, topics of great concern and political debate. We can decide whether zero population growth will be attained through social changes based on human choices or, instead, through increased mortality due to resource limitation, plagues, war, and environmental degradation.

CONCEPT CHECK 53.6

1. How does a human population's age structure affect its growth rate?
2. How have the rate and number of people added to the human population each year changed in recent decades?
3. **WHAT IF?** Type "personal ecological footprint calculator" into a search engine and use one of the resulting calculators to estimate your footprint. Is your current lifestyle sustainable? If not, what choices can you make to influence your own ecological footprint?

For suggested answers, see Appendix A.

53 Chapter Review



Go to **Mastering Biology** for Assignments, the eText, the Study Area, and Dynamic Study Modules.

SUMMARY OF KEY CONCEPTS

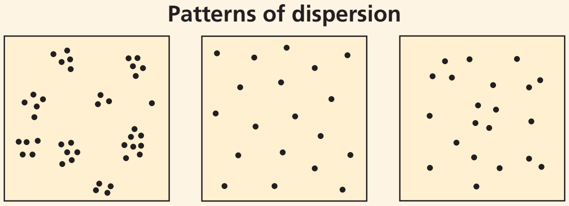
To review key terms, go to the **Vocabulary Self-Quiz** in the **Mastering Biology** eText or Study Area, or go to goo.gl/zkjz9t.

CONCEPT 53.1

Biotic and abiotic factors affect population density, dispersion, and demographics *(pp. 1191–1195)*

- Population **density**—the number of individuals per unit area or volume—reflects the interplay of births, deaths, immigration, and emigration. Environmental and social factors influence the **dispersion** of individuals.

Patterns of dispersion



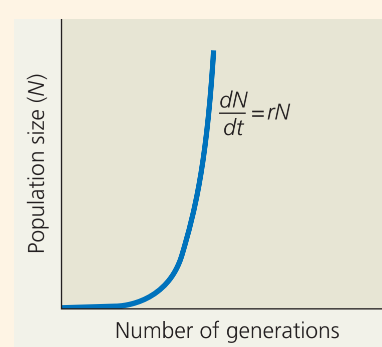
- Populations increase from births and **immigration** and decrease from deaths and **emigration. Life tables** and **survivorship curves** summarize specific trends in **demography**.

? *Gray whales (*Eschrichtius robustus*) gather each winter near Baja California to give birth. How might such behavior make it easier for ecologists to estimate birth and death rates for the species?*

CONCEPT 53.2

The exponential model describes population growth in an idealized, unlimited environment *(pp. 1196–1197)*

- If immigration and emigration are ignored, a population's per capita growth rate equals its birth rate minus its death rate.
- The **exponential growth** equation $dN/dt = rN$ represents a population's growth when resources are relatively abundant, where r is the **intrinsic rate of increase** and N is the number of individuals in the population.



? *Suppose one population has an r that is twice as large as the r of another population. What is the maximum size that both populations will reach over time, based on the exponential model?*